Pratik Pujari | CMPN B | 23102B0008

1. Which threshold value implements the desired fan-control rule?

- $\theta = 1$
- $\theta = 2$
- $\theta = 3$
- $\theta = 4$

2. For input (1, 0, 1, 1), what output does the controller produce?

- $y = 0$, because only one condition is absent
- $y = 1$, because three indicators are active
- $y = 0$, because all four indicators must be active
- $y = 1$, because the first indicator is active

3. How many binary input patterns switch on the fan?

- 5
- 4
- 6
- 8

4. If the threshold is accidentally changed to $\theta = 2$, what happens?

- The fan switches on less often than intended.
- The fan switches on when two or more indicators are active.
- The fan switches on only when all inputs are 1, making activation too strict.
- The fan never switches on.

5. Which property makes this controller a threshold-logic system?

- ( ) It predicts future classroom occupancy states.
- ( ) It updates weights after every incorrect output during a training process
- ( ) It requires a hidden layer to make a decision.
- (•) It compares a summed input value with a fixed threshold.

Case Study 2

Data for Questions 6 to 10

Scholarship Application Screening

A scholarship office builds a first-stage screening aid. The aid does not make the final award decision; it only flags applications that satisfy a simple rule for immediate review. Each application is represented by two binary features:

x_1 : academic score is above the cutoff, x_2 : financial-need certificate is verified.

The office wants the system to flag an application when at least one of these two conditions is true. A perceptron with weights w_1, w_2, bias b, and decision rule

$$\hat{y} = \begin{cases} 1, & \text{if } w_1x_1 + w_2x_2 + b \geq 0, \\ 0, & \text{otherwise} \end{cases}$$

is tested for this screening rule.

6. Which logical behavior is described by the desired screening rule?

- ( ) AND
- (•) XOR
- ( ) OR
- ( ) XNOR

7. Which parameter setting correctly implements this rule?

- ( ) $w_1 = 1, w_2 = 1, b = -1.5$
- (•) $w_1 = 1, w_2 = 1, b = -0.5$
- ( ) $w_1 = -1, w_2 = -1, b = 0.5$
- ( ) $w_1 = 1, w_2 = -1, b = 0$

8. For $w_1 = 1, w_2 = 1, b = -0.5$, what is the score for input (0, 1)?

- (•) 0.5

8. For $w_1 = 1, w_2 = 1, b = -0.5$, what is the score for input (0, 1)?

- (selected) 0.5
- -0.5
- 0
- 1.5

9. What is the decision boundary for the parameter setting $w_1 = 1, w_2 = 1, b = -0.5$?

- $x_1 - x_2 - 0.5 = 0$
- $x_1 + x_2 + 0.5 = 0$
- $2x_1 + x_2 = 0$
- (selected) $x_1 + x_2 - 0.5 = 0$

10. Why can a single perceptron represent this screening rule?

- (selected) The positive and negative cases are linearly separable.
- The rule requires remembering earlier applications before deciding the next output.
- The rule has no decision boundary.
- The output changes randomly for the same input.

11. Which logical behavior does the handover rule represent?

- AND
- OR
- XOR
- NAND

12. What output is assigned to the input pattern (1, 1)?

- $y = 1$, because one item has been returned
- $y = 0$, because exactly one item has not been returned
- $y = 1$, because both checks are complete
- $y = 0$, because both values are the same

13. Which geometric property prevents a single perceptron from representing this rule?

- The positive cases lie on opposite corners of the input square.
- The inputs are binary rather than real valued, so the rule cannot use a separator.
- The output has only two possible labels.
- The rule uses two input variables.

14. Which modified lab rule would be linearly separable?

- Flag exactly one returned item
- Flag when the two item statuses are different.
- Flag only when both items have been returned.
- Flag when neither or both items have been returned.

15. What does linear separability mean in this setting?

- Every input must have label 1 before a straight boundary can be checked.
- A line can separate the 1-labeled and 0-labeled inputs.

15. What does linear separability mean in this setting?

- ( ) Every input must have label 1 before a straight boundary can be checked.
- (●) A line can separate the 1-labeled and 0-labeled inputs.
- ( ) The perceptron must avoid using weights.
- ( ) The input values must be stored in a sequence.

Case Study 4

Data for Questions 16 to 20

Parcel Routing Decision Review

A courier service trains a perceptron to classify parcels into "express" and "standard" routing. Each parcel is described by two numerical features:

x_1 : compactness score, x_2 : priority-label score.

The label is $y = 1$ for express routing and $y = -1$ for standard routing. The perceptron has no bias term and uses the update rule

$w \leftarrow w + yx$

whenever $y(w^\top x) \leq 0$. The training samples are processed in the following order:

Parcel	x_1	x_2	y
P_1	2	1	1
P_2	1	2	1
P_3	−1	−1	−1

Training starts with $w = (0, 0)$.

16. What is the weight vector after processing P_1 once?

- (●) w = (2, 1)
- ( ) w = (1, 2)
- ( ) w = (−1, −1)
- ( ) w = (0, 0)

17. After the update from P_1, what is $\boldsymbol{w}^{\top}\boldsymbol{x}$ for P_2?

- 2
- 3
- 4
- 5

18. With $\boldsymbol{w} = (2, 1)$, is P_3 correctly classified?

- No, because the raw score $\boldsymbol{w}^{\top}\boldsymbol{x}$ is greater than zero
- No, because $y(\boldsymbol{w}^{\top}\boldsymbol{x}) < 0$
- Yes, because $\boldsymbol{w}^{\top}\boldsymbol{x} = 0$.
- Yes, because $y(\boldsymbol{w}^{\top}\boldsymbol{x}) > 0$.

19. Which condition is needed for the perceptron convergence guarantee?

- Every feature must be binary.
- The data must be linearly separable.
- The learning rate must be zero throughout all training updates.
- All labels must be positive.

20. Which vector separates the three listed parcels?

- $\boldsymbol{w} = (-1, 1)$
- $\boldsymbol{w} = (1, -1)$
- $\boldsymbol{w} = (-1, -1)$
- $\boldsymbol{w} = (1, 1)$